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RESEARCH ARTICLE

# Input and output constraints control strategy-based model predictive control for double-link overhead cranes

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**Abstract**
In this paper, we propose a novel control scheme that integrates a disturbance observer with a nonlinear anti-swing control approach for underactuated overhead crane systems. Unlike existing methods, this strategy accounts for double-pendulum dynamics, unknown disturbances, and system constraints. First, a high-gain observer is designed based on the system's dynamic equations to estimate the unknown disturbances. Then, a Lyapunov-based model predictive controller is developed, using the observer's information along with a second-order sliding mode controller, which is used for tracking while adhering to system constraints. Through comparative simulations with conventional controllers, the proposed method is validated, demonstrating improved performance and robustness.

**Keywords** Double-pendulum overhead cranes · Underactuated system · High-gain observer (HGO) · Second-order sliding mode control (SO-SMC) · Lyapunov-based model predictive control (LMPC)

## 1 Introduction

The overhead crane is one of the stationary cranes widely used in many different industries, such as maritime, construction, and manufacturing, due to the flexibility and outstanding load-lifting capacity of this device. This encourages scientists to investigate and develop this device further into a fully automatic device. However, this faces many difficulties, and one of the main reasons is that this system does not have enough actuators with fewer control inputs than the number of output signals [1]. In addition, the harsh working environment (dust, wind, etc.), the limitation of real actuators, and changes in system parameters also pose many challenges in both driving the load to the desired position and ensuring safe operation [2, 3]. Therefore, the development of automatic controllers for overhead crane systems is still a potential research area.

This type of crane is often combined with a trolley to move back and forth on the gantry. The load is usually attached to the trolley through a hook and cable. Currently, for the convenience of the research process, most publications in the past simplified the crane model by considering the cable and the load system as a single pendulum or the cable length as constant [4, 5]. With the aforementioned modeling simplifications, various advanced control strategies have been developed, including model predictive controllers [6], enhanced coupling controllers [7], adaptive hierarchical sliding mode controllers [8, 9], and neural network controllers [10], among others. These publications have largely addressed and resolved the challenges associated with crane control. However, despite their convenience in modeling the system, such assumptions have considerable limitations in real-world scenarios. In practice, the payload is suspended via a hook, which can result in double-pendulum dynamics rather than single-pendulum behavior. This more complex oscillatory motion significantly affects the swing behavior of

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the payload. Therefore, if we use a single-pendulum model, it may not be able to fully suppress the coupled oscillations in real-life systems, thereby degrading the swing suppression performance. Moreover, the cable length in actual crane operations is often time-varying due to hoisting and lowering actions, and assuming a cable length is fixed and constant may cause modeling inaccuracies, especially with substantial or rapid changes in length. All in all, these mismatches between model assumptions and actual system behavior can negatively impact both control accuracy and operational safety. Therefore, it is essential to take these elements into account while building a controller in order to preserve efficiency and safety while in operation.

Numerous control algorithms have been developed, primarily focusing on the double-pendulum effect and ignoring cable length fluctuations. The tracking and anti-vibration control issues for this system have been addressed by the conventional PID controller in [11, 12], which has continued to develop. In [11], Jaafar et al. combined the PID control algorithm with the PSO optimization algorithm to obtain the optimal parameter set for the controller. Along with that, [12] introduced a nonlinear PID algorithm to improve the ability to dominate trolley deviations compared to traditional PID sets. Besides the PID algorithm, sliding mode controller (SMC) [13–16] is also one of the methods that are still widely applied in controlling double-pendulum cranes thanks to its simplicity and ease of application. Among them, a typical one is a sliding mode controller combined with an extended state observer proposed in [16]. This approach simultaneously addresses trolley tracking control, mitigates load swing, and overcomes practical operational challenges by observing state variables and compensating for total disturbances. Many other advanced algorithms are also applied to this model, such as model predictive control [17], model predictive control based on flatness [18, 19], energy-optimal control integrating input constraints [20, 21]. Nevertheless, these studies have overlooked a crucial element: the variation in cable length, which plays a pivotal role in the process of load lifting and lowering.

At present, there are a limited number of studies that address both the double-pendulum phenomenon and the variation in cable length during operation concurrently [22–26]. Therefore, the authors in [22] suggest a nonlinear coupled-tracking anti-swing controller. This method uses a smooth tracking trajectory to minimize oscillations and enable stable system initialization. Additionally, [23] introduces a sophisticated adaptive fuzzy control rule to minimize the oscillations of the hook and the load by controlling the trolley's motion and cable length. To ensure efficient control, a fuzzy logic system (FLS) is used to compensate for the drive's dead zones. Through online estimation, [24] proposed an advanced adaptive controller for the double-pendulum overhead crane system, addressing uncertainties in payload mass. This approach enhances the system's resilience to uncertain changes. In particular, by adding more vibration data to the control procedure, the study improves vibration reduction performance. Furthermore, [25] suggests an adaptive fuzzy tracking controller that takes outside uncertainties and disruptions into consideration. Its two primary parts are a tracking signal designed to lessen system vibrations and enhance transient performance, and an adaptive law that estimates and compensates for friction, noise, and nonlinearity in the system using a radial basis function neural network (RBFNN). Finally, factors such as input constraints and swing angle limits in double-pendulum systems with variable cable lengths have garnered increasing attention, as noted in [26].

Based on a comprehensive analysis of recent literature on double-pendulum crane systems with variable cable lengths, it is evident that most existing control strategies are unable to effectively address multiple complex factors concurrently. These factors include swing angle limitation, input saturation, system parameter uncertainties, external disturbances, and output signal limitations. In particular, most studies do not consider the issue of output signal limitations, such as the velocity constraints of trolley and cable movement. This limitation has a direct impact on practical operation, as hardware systems can only function smoothly within a permissible range. Thus, this paper proposes a Lyapunov-based model predictive controller (LMPC) combined with a high-gain disturbance observer to comprehensively address the aforementioned issues for double-pendulum crane systems with variable cable lengths. The primary contributions of this paper can be summarized as follows:

- The proposed controller guarantees that the trolley and load are maneuvered to the desired trajectories while effectively minimizing load vibrations during movement, even under conditions where the system's input and output signals are constrained, which are not considered fully in previous studies [22–26].
- By employing on the high-gain disturbance observer (HGDO), this study also introduces an observation law designed to estimate and compensate for the cumulative effects of friction, external disturbances, and specific nonlinearities.
- In contrast to conventional MPC-based methods [17–19], the closed-loop stability of the entire system is more effectively ensured by integrating a candidate Lyapunov function condition derived from the second-order sliding mode control algorithm into the model predictive controller.

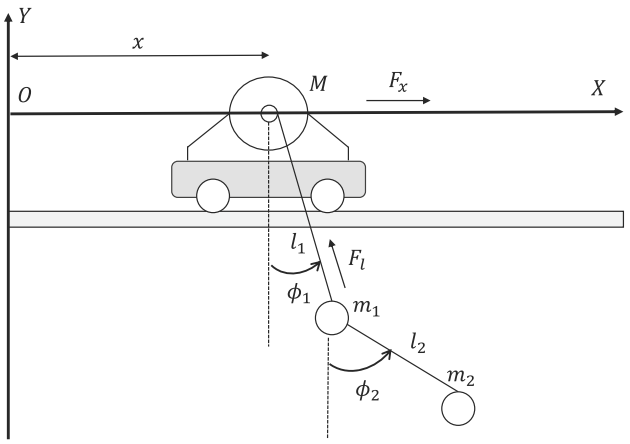



**Fig. 1** The 2D double-pendulum overhead crane

## 2 Mathematical model

Consider the motion of the 2D double-pendulum overhead crane (2DDOC) as illustrated in Fig. 1. The $OXY$ coordinate system is taken as the reference frame of the system. The crane model is described as follows: the payload is lifted by the slings and trusses of the crane, as shown in Fig. 1; the hoisting rope and sling are assumed to be massless and inelastic; the sling rope, denoted as $l_1$, is assumed to have a variable length. The model employed is the 2D model. The symbols in Fig. 1 refer to:

| | |
|---|---|
| $M$ | Mass of the cart |
| $m_1$ | Mass of the hook |
| $m_2$ | Mass of the payload |
| $x$ | Position of the cart in the $x$-axis |
| $l_1$ | Length of the wire from crane to hook |
| $l_2$ | Length of the wire from hook to payload |
| $\phi_1$ | Angle of rotation of the hook relative to the vertical |
| $\phi_2$ | Angle of rotation of the payload relative to the vertical |
| $F_x$ | Force applied to the cart |
| $F_l$ | Force applied to the rope $l_1$ |

Applying the Euler–Lagrange method as in [27], the crane system can be described as:

$$\boldsymbol{\mathcal{M}}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{\mathcal{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{\mathcal{G}}(\boldsymbol{q}) + \boldsymbol{\mathcal{D}} = \boldsymbol{\mathcal{U}} \tag{1}$$

in which $\boldsymbol{q} = \left[x, l_1, \phi_1, \phi_2\right]^{\mathrm{T}}$ is the the state vector and $\boldsymbol{\mathcal{U}} = \left[F_x, F_l, 0, 0\right]^{\mathrm{T}}$ denotes the driving force acting on the system. $\boldsymbol{\mathcal{M}}(\boldsymbol{q}) \in \mathbb{R}^{4,4}$ is the symmetric positive-definite inertia matrix; $\boldsymbol{\mathcal{C}}(\boldsymbol{q},\dot{\boldsymbol{q}}) \in \mathbb{R}^{4,4}$ presents the centrifugal and damping matrix; $\boldsymbol{\mathcal{G}}(\boldsymbol{q}) \in \mathbb{R}^{4,1}$ is the gravitational force vector and $\boldsymbol{\mathcal{D}} \in \mathbb{R}^{4,1}$ indicates the external disturbance vector.

The matrices coefficients of $\boldsymbol{\mathcal{M}}(\boldsymbol{q})$, $\boldsymbol{\mathcal{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})$, $\boldsymbol{\mathcal{G}}(\boldsymbol{q})$ are shown in Section 5.

## 3 Control structure formulation

In this section, we propose an adaptive model predictive control strategy for a double-pendulum crane system with variable cable length, as shown in Fig. 2. More specifically, first, a high-gain disturbance observer (HGDO) is initially designed to estimate external disturbances and components of uncertainty. Next, a second-order sliding surface is constructed, from which the corresponding Lyapunov candidate function is identified. The control signal is subsequently derived using a model predictive controller, where the computational model incorporates the total disturbance signal estimated by the observer. This process ensures compliance with both input–output signal constraints and the conditions imposed by the Lyapunov candidate function. As a result, the stability of the system is guaranteed.

### 3.1 High-gain disturbance observer

In practical applications, overhead crane systems are significantly influenced by external environmental factors, so the use of disturbance observers is crucial. Let $\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{M}}^{-1}(\boldsymbol{q})(\boldsymbol{\mathcal{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{\mathcal{G}}(\boldsymbol{q}) + \boldsymbol{\mathcal{D}}) = [\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4]^{\mathrm{T}}$ be the lump uncertainty that contains uncertainties and external disturbances. Then, Eq. (1) is written as follows:

$$\boldsymbol{\mathcal{M}}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{\mathcal{M}}(\boldsymbol{q})\boldsymbol{\mathcal{L}}(\boldsymbol{q},\dot{\boldsymbol{q}}) = \boldsymbol{\mathcal{U}} \tag{2}$$

Equation (2) is rewritten in the form of a system of state equations:

$$\begin{cases} \dot{\boldsymbol{x}}_1 = \boldsymbol{x}_2 \\ \dot{\boldsymbol{x}}_2 = \boldsymbol{f}(\boldsymbol{x},\boldsymbol{\mathcal{U}}) - \boldsymbol{\mathcal{L}} \\ \boldsymbol{y} = \boldsymbol{x}_1 \end{cases} \tag{3}$$

where $\boldsymbol{x}_1 = \boldsymbol{q}$ is the state vector, $\boldsymbol{x}_2 = \dot{\boldsymbol{q}}$ is the derivative of state vector, and $\boldsymbol{f}(\boldsymbol{x},\boldsymbol{\mathcal{U}}) = \boldsymbol{\mathcal{M}}^{-1}\boldsymbol{\mathcal{U}}$ is a nonlinear function.

Let $\widehat{\boldsymbol{\mathcal{L}}}$ be an estimate of $\boldsymbol{\mathcal{L}}$.

The high-gain observer is designed as follows:

$$\widehat{\boldsymbol{\mathcal{L}}} = \boldsymbol{z} - \frac{\boldsymbol{x}_2}{\varepsilon} \tag{4}$$

where $1/\varepsilon$ is an observer gain and $\boldsymbol{z}$ is an auxiliary state variable with the following dynamic:

$$\dot{\boldsymbol{z}} = \frac{1}{\varepsilon}\left(\boldsymbol{f}(\boldsymbol{x},\boldsymbol{\mathcal{U}}) + \frac{\boldsymbol{x}_2}{\varepsilon} - \boldsymbol{z}\right) \tag{5}$$

**Assumption 1** The derivative of the lumped disturbance $\boldsymbol{\mathcal{L}}$ is both continuous and bounded. As a result, there exists a positive constant $\delta$ that satisfies the following inequality: $\| \dot{\boldsymbol{\mathcal{L}}} \|_\infty \leqslant \delta$.

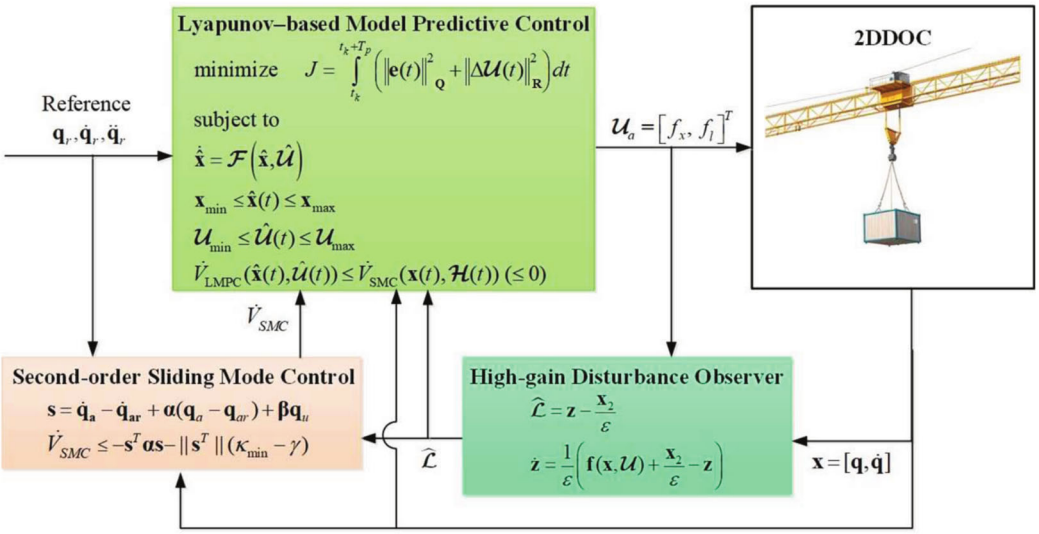



**Fig. 2** Scheme of closed-loop control

**Theorem 3.1** *With the observer suggested in Eq. (4), the estimation error* $\tilde{\mathcal{L}} = \mathcal{L} - \hat{\mathcal{L}}$ *reduces over time and enters a small neighborhood of origin when* $t \to \infty$*, depending on the chosen parameter* $\varepsilon$*. As a consequence,* $\hat{\mathcal{L}}$ *will converge to the small neighborhood of* $\mathcal{L}$*.*

***Proof*** Differentiating the auxiliary state variable $z$ in Eq. (4) with respect to time, we have:

$$\dot{\hat{\mathcal{L}}} = \dot{z} - \frac{\dot{\boldsymbol{x}}_2}{\varepsilon} \tag{6}$$

Let $\tilde{\mathcal{L}} = \mathcal{L} - \hat{\mathcal{L}}$ be the estimation error between the real disturbance $\mathcal{L}$ and the estimated value $\hat{\mathcal{L}}$. By substituting (6) into the disturbance estimation error equation and using equations (3), (4) and (5), its derivative is determined, as follows:

$$\begin{aligned}
\dot{\tilde{\mathcal{L}}} &= \dot{\mathcal{L}} - \dot{\hat{\mathcal{L}}} \\
&= \dot{\mathcal{L}} - \left( \dot{z} - \frac{\dot{\boldsymbol{x}}_2}{\varepsilon} \right) \\
&= \dot{\mathcal{L}} - \frac{1}{\varepsilon}\left( f(\boldsymbol{x}, \mathcal{U}) + \frac{\boldsymbol{x}_2}{\varepsilon} - z \right) + \frac{1}{\varepsilon}\left( f(\boldsymbol{x}, \mathcal{U}) - \mathcal{L} \right) \\
&= \dot{\mathcal{L}} - \frac{\hat{\mathcal{L}}}{\varepsilon} + \frac{\mathcal{L}}{\varepsilon}
\end{aligned} \tag{7}$$

which means that

$$\dot{\tilde{\mathcal{L}}} + \frac{1}{\varepsilon}\tilde{\mathcal{L}} = \dot{\mathcal{L}} \tag{8}$$

This is a first-order differential equation; solving this differential equation, we have:

$$\tilde{\mathcal{L}}_j = e^{(-1/\varepsilon)t}\tilde{\mathcal{L}}_j(0) + \varepsilon \dot{\mathcal{L}}_j, \ j = 1 \div 4 \tag{9}$$

with $\tilde{\mathcal{L}}_j(0)$ is the initial value of $\tilde{\mathcal{L}}_j$. Combined with Assumption 1, the following result is obtained:

$$|\tilde{\mathcal{L}}_j| \le e^{(-1/\varepsilon)t}|\tilde{\mathcal{L}}_j(0)| + \varepsilon\delta, \ j = 1 \div 4 \tag{10}$$

From Eq. (10), it can be seen that by appropriately choosing a very small parameter $\varepsilon$, the upper bound of $|\tilde{\mathcal{L}}_j|$ can be made arbitrarily small. This implies that the lumped disturbance estimate $\hat{\mathcal{L}}$ converges to the neighborhood of $\mathcal{L}$, completing the proof of Theorem 3.1. □

### 3.2 Second-order sliding mode control

The primary objective of this paper is to determine the control input $\mathcal{U}_a = [F_x, F_l]^{\mathrm{T}}$ such that the state variables reach their reference values $\boldsymbol{q}_r = [x_r, l_{1r}, \phi_{1r}, \phi_{2r}]^{\mathrm{T}}$. Given the underactuated nature of the system, the dynamics are divided into two subsystems: one for the actuated states $\boldsymbol{q}_a = [x, l_1]^{\mathrm{T}}$ and the other corresponding to the underactuated states $\boldsymbol{q}_u = [\phi_1, \phi_2]^{\mathrm{T}}$ as follows:

$$\begin{cases}
\mathcal{M}_{aa}(\boldsymbol{q})\ddot{\boldsymbol{q}}_a + \mathcal{M}_{au}(\boldsymbol{q})\ddot{\boldsymbol{q}}_u + \mathcal{M}_{aa}(\boldsymbol{q})\mathcal{L}_a(\boldsymbol{q}, \dot{\boldsymbol{q}}) \\
+ \mathcal{M}_{au}(\boldsymbol{q})\mathcal{L}_u(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \mathcal{U}_a \\
\mathcal{M}_{ua}(\boldsymbol{q})\ddot{\boldsymbol{q}}_a + \mathcal{M}_{uu}(\boldsymbol{q})\ddot{\boldsymbol{q}}_u + \mathcal{M}_{ua}(\boldsymbol{q})\mathcal{L}_a(\boldsymbol{q}, \dot{\boldsymbol{q}}) \\
+ \mathcal{M}_{uu}(\boldsymbol{q})\mathcal{L}_u(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \boldsymbol{0}
\end{cases} \tag{11}$$

where $\mathcal{L}_a(\boldsymbol{q}, \dot{\boldsymbol{q}}) = [\mathcal{L}_1, \mathcal{L}_2]^{\mathrm{T}}$, $\mathcal{L}_u(\boldsymbol{q}, \dot{\boldsymbol{q}}) = [\mathcal{L}_3, \mathcal{L}_4]^{\mathrm{T}}$ and the coefficients of matrices $\mathcal{M}_{aa}(\boldsymbol{q})$, $\mathcal{M}_{au}(\boldsymbol{q})$, $\mathcal{M}_{ua}(\boldsymbol{q})$, $\mathcal{M}_{uu}(\boldsymbol{q})$ are shown in Appendix.

From (11), we obtain the equivalent kinetic model of the actuated system:

$$\mathcal{M}_a(\boldsymbol{q})\ddot{\boldsymbol{q}}_a + \mathcal{M}_a(\boldsymbol{q})\mathcal{L}_a(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \mathcal{U}_a \tag{12}$$

where $\mathcal{M}_a = \mathcal{M}_{aa} - \mathcal{M}_{au}\mathcal{M}_{uu}^{-1}\mathcal{M}_{ua}$.

The control methods will be applied to (12) to determine the appropriate control signals, denoted by $\mathcal{U}_a$. Let $\tilde{\boldsymbol{q}}_a = \boldsymbol{q}_a - \boldsymbol{q}_{ar}$ be the tracking error vector and $\tilde{\boldsymbol{q}}_u = \boldsymbol{q}_u - \boldsymbol{q}_{ur}$ be the vibrant error vector, with $\boldsymbol{q}_{ar} = [x_r, l_{1r}]^{\mathrm{T}}$, $\boldsymbol{q}_{ur} = [\phi_{1r}, \phi_{2r}]^{\mathrm{T}} = [0, 0]^{\mathrm{T}}$. According to [28], the sliding surface is proposed as follows:

$$\boldsymbol{s} = \dot{\tilde{\boldsymbol{q}}}_a + \boldsymbol{\alpha}\tilde{\boldsymbol{q}}_a + \boldsymbol{\beta}\tilde{\boldsymbol{q}}_u = \dot{\boldsymbol{q}}_a - \dot{\boldsymbol{q}}_{ar} + \boldsymbol{\alpha}(\boldsymbol{q}_a - \boldsymbol{q}_{ar}) + \boldsymbol{\beta}\boldsymbol{q}_u \tag{13}$$

where $\boldsymbol{\alpha} = \mathrm{diag}(\alpha_1, \alpha_2)$ and $\boldsymbol{\beta} = \mathrm{diag}(\beta_1, \beta_2)$ with $\alpha_i > 0, \beta_i \in \mathbb{R}, i = 1, 2$.

The sliding surface $\boldsymbol{s}$ can be differentiated with respect to time as follows:

$$\begin{aligned}\dot{\boldsymbol{s}} &= \ddot{\tilde{\boldsymbol{q}}}_a + \boldsymbol{\alpha}\dot{\tilde{\boldsymbol{q}}}_a + \boldsymbol{\beta}\dot{\tilde{\boldsymbol{q}}}_u \\ &= \ddot{\boldsymbol{q}}_a - \ddot{\boldsymbol{q}}_{ar} + \boldsymbol{\alpha}(\dot{\boldsymbol{q}}_a - \dot{\boldsymbol{q}}_{ar}) + \boldsymbol{\beta}\dot{\boldsymbol{q}}_u\end{aligned} \tag{14}$$

Substituting $\ddot{\boldsymbol{q}}_a$ from (12) into (14), the equation (14) is rewritten as follows:

$$\begin{aligned}\dot{\boldsymbol{s}} = \mathcal{M}_a^{-1}(\boldsymbol{q})\mathcal{U}_a - \mathcal{L}_a(\boldsymbol{q}, \dot{\boldsymbol{q}}) \\ - \ddot{\boldsymbol{q}}_{ar} + \boldsymbol{\alpha}(\dot{\boldsymbol{q}}_a - \dot{\boldsymbol{q}}_{ar}) + \boldsymbol{\beta}\dot{\boldsymbol{q}}_u\end{aligned} \tag{15}$$

The controller is designed such that the sliding surface $\boldsymbol{s}$ converges to zero as $t$ approaches infinity. The controller is proposed as follows:

$$\mathcal{U}_a = \mathcal{U}_{eq} + \mathcal{U}_{sw} \tag{16}$$

in which

$$\begin{aligned}\mathcal{U}_{eq} &= \mathcal{M}_a(\boldsymbol{q})\widehat{\mathcal{L}_a} - \mathcal{M}_a \\ &\quad \left[2\boldsymbol{\alpha}\dot{\tilde{\boldsymbol{q}}}_a + \boldsymbol{\beta}\dot{\tilde{\boldsymbol{q}}}_u + \boldsymbol{\alpha}\boldsymbol{\alpha}\tilde{\boldsymbol{q}}_a + \boldsymbol{\alpha}\boldsymbol{\beta}\tilde{\boldsymbol{q}}_u - \ddot{\boldsymbol{q}}_{ar}\right] \\ \mathcal{U}_{sw} &= -\mathcal{M}_a\boldsymbol{\kappa}\mathrm{sign}(\boldsymbol{s})\end{aligned}$$

where $\boldsymbol{\kappa} = \mathrm{diag}(\kappa_1, \kappa_2)\mathrm{with}\kappa_i > 0, i = 1, 2$.

Substituting $\mathcal{U}_a$ from (16) into (15), the derivative of the sliding surface becomes:

$$\begin{aligned}\dot{\boldsymbol{s}} &= \widehat{\mathcal{L}_a} - \mathcal{L}_a - \boldsymbol{\alpha}\left(\ddot{\tilde{\boldsymbol{q}}}_a + \boldsymbol{\alpha}\dot{\tilde{\boldsymbol{q}}}_a + \boldsymbol{\beta}\dot{\tilde{\boldsymbol{q}}}_u\right) - \boldsymbol{\kappa}\mathrm{sign}(s) \\ &= -\boldsymbol{\alpha}\boldsymbol{s} - \boldsymbol{\kappa}\mathrm{sign}(s) - \tilde{\mathcal{L}}_a\end{aligned} \tag{17}$$

**Theorem 3.2** *The system will exhibit asymptotic stability if the controller is designed according to* (16)*. Consequently, the tracking errors $\tilde{\boldsymbol{q}}_a$ and $\tilde{\boldsymbol{q}}_u$ will converge to zero, and the system's trajectory will approach the desired path.*

***Proof*** Consider the Lyapunov candidate function:

$$V_{\mathrm{SMC}} = \frac{1}{2}\boldsymbol{s}^{\mathrm{T}}\boldsymbol{s} \tag{18}$$

Substituting (16) into (15), then the time derivative of the Lyapunov function is obtained:

$$\begin{aligned}\dot{V}_{\mathrm{SMC}} &= \boldsymbol{s}^{\mathrm{T}}\dot{\boldsymbol{s}} \\ &= \boldsymbol{s}^{\mathrm{T}}\left(-\boldsymbol{\alpha}\boldsymbol{s} - \boldsymbol{\kappa}\mathrm{sign}(\boldsymbol{s}) - \tilde{\mathcal{L}}_a)\right) \\ &= -\boldsymbol{s}^{\mathrm{T}}\boldsymbol{\alpha}\boldsymbol{s} - \boldsymbol{s}^{\mathrm{T}}\boldsymbol{\kappa}\mathrm{sign}(\boldsymbol{s}) - \boldsymbol{s}^{\mathrm{T}}\tilde{\mathcal{L}}_a\end{aligned} \tag{19}$$

According to Theorem 3.1, $\| \tilde{\mathcal{L}}_a \|$ is bounded. As a result, there exists a positive constant $\gamma$ that satisfies the following inequality:$\| \tilde{\mathcal{L}}_a \| \leq \gamma$. The control parameter $\kappa_1, \kappa_2$ is chosen such that $\kappa_{\min} \geq \gamma$, where $\kappa_{\min} = \min(\kappa_1, \kappa_2)$. Then, equation (19) can be rewritten as

$$\dot{V}_{\mathrm{SMC}} \leqslant -\boldsymbol{s}^{\mathrm{T}}\boldsymbol{\alpha}\boldsymbol{s} - \| \boldsymbol{s}^{\mathrm{T}} \| (\kappa_{\min} - \gamma) \leqslant 0 \tag{20}$$

Thus, by Barbalat's lemma, $\lim_{t\to\infty} V_{\mathrm{SMC}} = 0$, resulting in $\lim_{t\to\infty} \boldsymbol{s} = \boldsymbol{0}$.

In physical terms, gravity acts to return the hook and the payload to their equilibrium position, which corresponds to $\phi_1 \to 0$ and $\phi_2 \to 0$. As $\phi_1$ and $\phi_2$ remain bounded, the tracking error converges to the origin with $\alpha_i > 0$ and arbitrary gain $\beta_i$. In summary, the second-order sliding mode control structure in (13) ensures that all system states converge to the desired values. Thus, the proof of Theorem 3.2 is completed. □

### 3.3 Lyapunov-based model predictive control

The crane system's structural intricacies pose significant challenges in practical applications. Beyond the fundamental tasks of trolley control and mitigating load oscillations, it is essential to rigorously impose constraints on both state variables and control inputs. This necessity arises from the fact that actuators may not consistently respond in accordance with the controller's commands. Therefore, we need to have an effective control strategy in order to precisely control state variables like trolley speed and load swing angle, as well as satisfy system outputs constraints while preserving overall system stability. The LMPC method enhances system stability by developing the principles of MPC. LMPC takes advantage of traditional MPC and ensures closed-loop stability through a Lyapunov function using the SO-SMC information. This approach provides accurate trajectory tracking for state variables and well-regulated control inputs and secures system stability via the auxiliary Lyapunov function.

Let $\boldsymbol{x} = [\boldsymbol{x}_1, \boldsymbol{x}_2]^{\mathrm{T}}$ with $\boldsymbol{x}_1 = \boldsymbol{q}$ and $\boldsymbol{x}_2 = \dot{\boldsymbol{q}}$, the system used to design the LMPC method after estimating lumped disturbance becomes:

$$\dot{\boldsymbol{x}} = \begin{bmatrix} \dot{\boldsymbol{q}} \\ \boldsymbol{\mathcal{M}}(\boldsymbol{q})^{-1}\boldsymbol{\mathcal{U}} - \widehat{\boldsymbol{\mathcal{L}}} \end{bmatrix} = \boldsymbol{\mathcal{F}}(\boldsymbol{x}, \boldsymbol{\mathcal{U}}) \tag{21}$$

Consider a prediction horizon depicted by $T_p = N_p T_s$, where $N_p$ indicates the length of the prediction horizon and $T_s$ refers to the sampling time. The value of $N_p$ is chosen based on the balance between the ability to predict and the computational complexity of the system. A larger $N_p$ generally improves prediction accuracy and control performance, but also increases computational burden. The cost function and the constraints for the LMPC controller are constructed as follows:

$$J = \int_{t_k}^{t_k+T_p} \left( \left\| \boldsymbol{e}(t) \right\|_{\boldsymbol{Q}}^2 + \left\| \Delta\boldsymbol{\mathcal{U}}(t) \right\|_{\boldsymbol{R}}^2 \right) \mathrm{d}t \tag{22}$$

$$\begin{aligned} \text{subject to } \dot{\widehat{\boldsymbol{x}}} &= \boldsymbol{\mathcal{F}}\left(\widehat{\boldsymbol{x}}, \widehat{\boldsymbol{\mathcal{U}}}\right) \\ \boldsymbol{x}_{\min} &\leqslant \widehat{\boldsymbol{x}}(t) \leqslant \boldsymbol{x}_{\max} \\ \boldsymbol{\mathcal{U}}_{\min} &\leqslant \widehat{\boldsymbol{\mathcal{U}}}(t) \leqslant \boldsymbol{\mathcal{U}}_{\max} \\ \dot{V}_{\mathrm{LMPC}}(\widehat{\boldsymbol{x}}(t), \widehat{\boldsymbol{\mathcal{U}}}(t)) &\leqslant \dot{V}_{\mathrm{SMC}}(\boldsymbol{x}(t), \boldsymbol{\mathcal{H}}(t)) \end{aligned}$$

in which $\widehat{\boldsymbol{x}}_1(t)$ and $\widehat{\boldsymbol{\mathcal{U}}}(t)$ are the predicted state and the predicted output, respectively; $\boldsymbol{e}(t) = \widehat{\boldsymbol{x}}_1(t) - \boldsymbol{q}_r(t)$ is the state tracking error at time t; $\Delta\boldsymbol{\mathcal{U}}(t)$ represents the deviation between the control input at two consecutive sampling time; $\boldsymbol{Q} \in \mathrm{R}^{4\times4}$ and $\boldsymbol{R} \in \mathrm{R}^{2\times2}$ are positive-definite weighted matrices; $\dot{V}_{\mathrm{SMC}}(\boldsymbol{x}(t), \boldsymbol{\mathcal{H}}(t))$ is the Lyapunov function calculated based on SO-SMC as in Section 3.2 with $\boldsymbol{\mathcal{H}}(t)$ is the control vector employed as follows:

$$\begin{aligned} \boldsymbol{\mathcal{H}} = \boldsymbol{\mathcal{M}}_a(\boldsymbol{q})\,\widehat{\boldsymbol{\mathcal{L}}_a} - \boldsymbol{\mathcal{M}}_a \Big[ & 2\boldsymbol{\alpha}\dot{\tilde{\boldsymbol{q}}}_a + \boldsymbol{\beta}\ddot{\tilde{\boldsymbol{q}}}_u \\ & + \boldsymbol{\alpha}^{\mathrm{T}}\boldsymbol{\alpha}\tilde{\boldsymbol{q}}_a + \boldsymbol{\alpha}\boldsymbol{\beta}\tilde{\boldsymbol{q}}_u - \ddot{\boldsymbol{q}}_{ar} \Big] - \boldsymbol{\mathcal{M}}_a \boldsymbol{\kappa}\,\mathrm{sign}(\boldsymbol{s}) \end{aligned} \tag{23}$$

These above constraints ensure that the 2DDOC functions within its dynamic limits, keeping the system states and control inputs remain the permissible operating range of the actuators. The final constraint ensures that the derivative of the Lyapunov function $\dot{V}_{\mathrm{LMPC}}(\widehat{\boldsymbol{x}}(t), \widehat{\boldsymbol{\mathcal{U}}}(t))$ (explained in (28)) for the control signal to be constructed is always less than or equal to the derivative of the Lyapunov function $\dot{V}_{\mathrm{SMC}}(\boldsymbol{x}(t), \boldsymbol{\mathcal{H}}(t))$. This enhances the stability of the LMPC controller compared to the traditional MPC controller.

The controller used the sample time $T_s$ to solve the loop, in each loop the controller minimizes the cost function $J$ to archive control signals $\boldsymbol{\mathcal{U}}$. After each cycle, the optimal solution is determined:

$$\boldsymbol{\mu}^*(t_k) = \mathrm{argmin} J(t_k)$$

with $\boldsymbol{\mu}(t_k) = \left[\widehat{\boldsymbol{\mathcal{U}}}(t_k), \widehat{\boldsymbol{\mathcal{U}}}(t_k + T_s), ..., \widehat{\boldsymbol{\mathcal{U}}}\left(t_k + (N_p - 1)T_s\right)\right]$. The first value $\widehat{\boldsymbol{\mathcal{U}}}(t_k)$ of the vector $\boldsymbol{\mu}^*$ is taken as the control signal at the $t_k$ period. This process will be continued in the next computation cycles. The specifics of the complete closed-loop control system are outlined in Algorithm 1.

In the experiment, the process of optimizing the above function $J(t_k)$ is performed on the integrated calculator *Nonlinear MPC* of MATLAB/Simulink software.

**Algorithm 1**:
1. Measure current states $\boldsymbol{q}(t_k)$.
2. Provide the current state $\boldsymbol{q}(t_k)$ to the HGDO for estimating the lumped disturbances $\boldsymbol{\mathcal{L}}(t_k)$. Then, the estimated lumped disturbances will be used in the LMPC controller with $\widehat{\boldsymbol{\mathcal{L}}}(t_k)$ at time step $t_k$ and the SO-SMC controller with $\widehat{\boldsymbol{\mathcal{L}}}(t_k)$.
3. The SO-SMC calculates the derivative of the Lyapunov function $\dot{V}_{\mathrm{SMC}}(\boldsymbol{x}(t_k), \boldsymbol{\mathcal{H}}(t_k))$
4. Provide the derivative of the Lyapunov function $\dot{V}_{\mathrm{SMC}}(\boldsymbol{x}(t_k), \boldsymbol{\mathcal{H}}(t_k))$ at time step $t_k$ to the LMPC for the computation of the cost function $J$ and its associated constraints. The solution to the LMPC problem is provided in (22), taking into account the given constraints to derive the (sub-)optimal control input $\boldsymbol{\mathcal{U}}(t_k)$.
5. Implement the 2DDOC system using the control input $\boldsymbol{\mathcal{U}}(t_k)$.
6. At the next sampling instant, increment the time variable $t_k$ by $T_s$, and return to step 1 to repeat the process.

**Theorem 3.3** *Based on Theorems 3.1 and 3.2, the observer error converges to a small neighborhood of origin when $t \to \infty$, depending on the chosen parameter $\varepsilon$. Furthermore, the derivative of the Lyapunov function satisfies $\dot{V}_{SMC} < 0$. As a result, the solution of the closed-loop system under Algorithm 1 of the LMPC demonstrates asymptotic stability. This guarantees that the trajectory of the 2DDOC system converges to the desired path through the LMPC approach.*

***Proof*** The sliding surface for the LMPC controller is proposed as follows:

$$\boldsymbol{s}_L = \dot{\widehat{\boldsymbol{q}}}_a - \dot{\boldsymbol{q}}_{ar} + \boldsymbol{\alpha}(\widehat{\boldsymbol{q}}_a - \boldsymbol{q}_{ar}) + \boldsymbol{\beta}\widehat{\boldsymbol{q}}_u \tag{24}$$

with $\hat{\boldsymbol{q}}_a = [\hat{x}, \hat{l}_1]^{\mathrm{T}}$, $\widehat{\boldsymbol{q}}_a = [\widehat{x}, \widehat{l}_1]^{\mathrm{T}}$, and $\widehat{\boldsymbol{q}}_u = [\widehat{\phi}_1, \widehat{\phi}_2]^{\mathrm{T}}$ are the values estimated by the MPC.

The time derivative of the sliding surface $\boldsymbol{s}_L$ is as follows:

$$\begin{aligned} \dot{\boldsymbol{s}}_L = {} & \boldsymbol{\mathcal{M}}_a^{-1}(\boldsymbol{q})\,\widehat{\boldsymbol{\mathcal{U}}}_a - \widehat{\boldsymbol{\mathcal{L}}}_a(\boldsymbol{q}, \dot{\boldsymbol{q}}) - \ddot{\boldsymbol{q}}_{ar} + \boldsymbol{\alpha}(\dot{\widehat{\boldsymbol{q}}}_a - \dot{\boldsymbol{q}}_{ar}) \\ & + \boldsymbol{\beta}\ddot{\widehat{\boldsymbol{q}}}_u \end{aligned} \tag{25}$$

Employing the disturbance estimation using the high-gain disturbance observer (HGDO), a dependent Lyapunov-based function is employed as follows:

$$\boldsymbol{\mathcal{H}} = \boldsymbol{\mathcal{M}}_a(\boldsymbol{q})\,\widehat{\boldsymbol{\mathcal{L}}_a} - \boldsymbol{\mathcal{M}}_a \Big[ 2\boldsymbol{\alpha}\dot{\tilde{\boldsymbol{q}}}_a + \boldsymbol{\beta}\ddot{\tilde{\boldsymbol{q}}}_u$$

$$+\boldsymbol{\alpha}^{\mathrm{T}}\boldsymbol{\alpha}\tilde{\boldsymbol{q}}_a+\boldsymbol{\alpha}\boldsymbol{\beta}\tilde{\boldsymbol{q}}_u-\ddot{\boldsymbol{q}}_{ar}\Big]-\boldsymbol{\mathcal{M}}_a\boldsymbol{\kappa}\mathrm{sign}(\boldsymbol{s}_L) \tag{26}$$

The Lyapunov function for the LMPC is selected as follows:

$$V_{\mathrm{LMPC}}=\frac{1}{2}\boldsymbol{s}_L^{\mathrm{T}}\boldsymbol{s}_L \tag{27}$$

Differentiating the Lyapunov function (27) and then substituting (25), we get:

$$\begin{aligned}&\dot{V}_{\mathrm{LMPC}}\\&=\boldsymbol{s}_L^{\mathrm{T}}\dot{\boldsymbol{s}}_L\\&=\boldsymbol{s}_L^{\mathrm{T}}\left[\boldsymbol{\mathcal{M}}_a^{-1}(\boldsymbol{q})\widehat{\boldsymbol{\mathcal{U}}}_a-\widehat{\boldsymbol{\mathcal{L}}}_a(\boldsymbol{q},\dot{\boldsymbol{q}})-\ddot{\boldsymbol{q}}_{ar}+\boldsymbol{\alpha}(\dot{\tilde{\boldsymbol{q}}}_a-\dot{\boldsymbol{q}}_{ar})+\boldsymbol{\beta}\dot{\tilde{\boldsymbol{q}}}_u\right]\\&\leq\boldsymbol{s}_L^{\mathrm{T}}\left[\boldsymbol{\mathcal{M}}_a^{-1}(\boldsymbol{q})\boldsymbol{\mathcal{H}}-\boldsymbol{\mathcal{L}}_a(\boldsymbol{q},\dot{\boldsymbol{q}})-\ddot{\boldsymbol{q}}_{ar}+\boldsymbol{\alpha}(\dot{\boldsymbol{q}}_a-\dot{\boldsymbol{q}}_{ar})+\boldsymbol{\beta}\dot{\boldsymbol{q}}_u\right]\end{aligned} \tag{28}$$

We can infer that:

$$\dot{V}_{\mathrm{LMPC}}\leq\dot{V}_{\mathrm{SMC}}\leq-\boldsymbol{s}_L^{\mathrm{T}}\boldsymbol{\alpha}\boldsymbol{s}-\parallel\boldsymbol{s}_L^{\mathrm{T}}\parallel(\kappa_{\min}-\gamma)\leqslant 0 \tag{29}$$

As a result, the closed-loop system governed by LMPC Algorithm 1 demonstrates asymptotic stability, guaranteeing that the 2DDOC system's trajectory follows the reference trajectory. □

## 4 Numerical results

This section will present simulations performed in MATLAB/SIMULINK to evaluate the effectiveness of the proposed control scheme. The initial values of the states are set as $\boldsymbol{q}(0)=[0.5,0.5,0.02,0.02]^{\mathrm{T}}$, and the observer are $\widehat{\boldsymbol{\mathcal{L}}}(0)=[0,0,0,0]^{\mathrm{T}}$.

The reference trajectory is chosen as follows:

$$x_r(t)\ (\mathrm{m})=\begin{cases}0.5+0.6\mathrm{pimf}(t,[0,10,15,25]), & \text{if } t<25\\0.5+0.5\mathrm{pimf}(t,[30,40,45,55]), & \text{otherwise}\end{cases}$$

$$l_{1r}(t)\ (\mathrm{m})=\begin{cases}0.5+0.5\mathrm{pimf}(t,[0,10,15,25]), & \text{if } t<25\\0.5+0.4\mathrm{pimf}(t,[30,40,45,55]), & \text{otherwise}\end{cases}$$

$$\phi_{1r}(t)=\phi_{2r}(t)=0\ (\mathrm{rad})$$

To highlight the advances of the suggested tracking approach, some comparison results with the SMC-HG (sliding mode control based on the high-gain disturbance observer), MPC-HG (model predictive control based on the high-gain disturbance observer), and the PID methods are provided.

1. For the SMC-HG method, the controller is designed as in (16).
2. The cost function and the constraints for the MPC-HG controller are determined:

$$J=\int_{t_k}^{t_k+T_p}\left(\left\|\boldsymbol{e}(t)\right\|_{\boldsymbol{Q}}^2+\left\|\Delta\boldsymbol{\mathcal{U}}(t)\right\|_{\boldsymbol{R}}^2\right)\mathrm{d}t \tag{30}$$

$$\begin{aligned}\text{subject to: }&\dot{\hat{\boldsymbol{x}}}=\boldsymbol{\mathcal{F}}(\hat{\boldsymbol{x}},\widehat{\boldsymbol{\mathcal{U}}})\\&\boldsymbol{x}_{\min}\leqslant\hat{\boldsymbol{x}}(t)\leqslant\boldsymbol{x}_{\max}\\&\boldsymbol{\mathcal{U}}_{\min}\leqslant\widehat{\boldsymbol{\mathcal{U}}}(t)\leqslant\boldsymbol{\mathcal{U}}_{\max}\end{aligned}$$

In which the notations used are consistent with those presented in (22).
3. For the PID controller, the control signals are calculated based on:

$$\begin{aligned}\boldsymbol{\mathcal{U}}_a=&K_{P1}\boldsymbol{e}_a+K_{I1}\int\boldsymbol{e}_a\mathrm{d}t+K_{D1}\frac{\mathrm{d}\boldsymbol{e}_a}{\mathrm{d}t}+K_{P2}\boldsymbol{e}_u\\&+K_{I2}\int\boldsymbol{e}_u\mathrm{d}t+K_{D2}\frac{\mathrm{d}\boldsymbol{e}_u}{\mathrm{d}t}\end{aligned} \tag{31}$$

where $\boldsymbol{e}_a=[x_r-x,l_{1r}-l_1]^{\mathrm{T}}$ and $\boldsymbol{e}_u=[\phi_{1r}-\phi_1,\phi_{2r}-\phi_2]^{\mathrm{T}}$.

The parameters of the 2DDOC model and the controllers are outlined in Table 1. Note that the SMC-HG, MPC-HG, and observer parameters used in the comparison are identical to those in the proposed algorithm. An important point to consider is that the parameters of the controllers and observer are selected to satisfy the conditions given in equations (6), (13), (16), and (22). To achieve this, the procedure is carried out as follows:

- Step 1: Model the 2DDOC system mathematically using MATLAB/SIMULINK.
- Step 2: Implement the block corresponding to the HGDO and choose the coefficient $\varepsilon$ to be sufficiently small.
- Step 3: Adjust the parameters of the SO-SMC using the signal $\widehat{\boldsymbol{\mathcal{L}}}$ generated by the observer, ensuring that the matrices $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, and $\boldsymbol{\kappa}$ meet the requirements specified in Equations (13) and (16).
- Step 4: Design the LMPC controller using the signal $\dot{V}_{\mathrm{SMC}}$ and the estimated signals from the HGDO. The coefficients of the $\boldsymbol{Q}$ and $\boldsymbol{R}$ matrices are selected based on the system's control objectives, with an emphasis on orbit-tracking accuracy and vibration suppression.
- Step 5: Finally, validate the results. If the system fails to track the reference trajectory or exhibits excessive vibration, return to Step 1 and adjust the controller parameters accordingly.

**Table 1** Simulation parameters

| 2DDOC | SMC-HG | MPC | PID |
|---|---|---|---|
| $M = 5\text{kg}$ | $\alpha_1 = 30$ | $T_s = 0.05$, $N_p = 10$ | $K_{P1} = 262.41$ |
| $m_1 = 2\text{kg}$ | $\alpha_2 = 30$ | $\boldsymbol{\mathcal{U}}_{\max} = [20; 60]$ | $K_{I1} = 8.382$ |
| $m_2 = 2.5\text{kg}$ | $\beta_1 = -6$ | $\boldsymbol{\mathcal{U}}_{\min} = [-20; -60]$ | $K_{D1} = 1825.2$ |
| $l_2 = 0.2\text{m}$ | $\beta_2 = -6$ | $\boldsymbol{x}_{\max} = [1.2; 1.2; 0.08; 0.08]$ | $K_{P2} = 1.7494$ |
| $g = 9.81\text{m/s}^2$ | $\kappa_1 = 0.8$ | $\boldsymbol{x}_{\min} = [0; 0; -0.08; -0.08]$ | $K_{I2} = 0.0559$ |
| $f_x = 82\text{Nm/s}$ | $\kappa_2 = 0.8$ | $\boldsymbol{Q} = \text{diag}(2000, 1000, 20, 20)$ | $K_{D2} = 12.168$ |
| $f_l = 75\text{Nm/s}$ | $\varepsilon = 0.001$ | $\boldsymbol{R} = \text{diag}(1, 1)$ | |

This study validates the superiority and robustness of the proposed controller through two simulation scenarios under different disturbance conditions.

### 4.1 Scenario 1

This simulation scenario is implemented with the disturbance $\boldsymbol{\mathcal{D}} = [9\sin(0.02t), 39\cos(0.01t), 0.15\sin(t), 0.25\sin(0.5t)]^{\mathrm{T}}$.

Figure 3 and Fig. 4 show the tracking and anti-vibration results of four controllers. These figures show that the proposed LMPC-HG controller demonstrates clear advantages in both tracking performance and vibration suppression when compared to the PID, SMC-HG, and MPC-HG control schemes. The LMPC-HG controller provides improved tracking precision in Fig. 3, which displays the output signals for the trolley displacement $x$ and hook length $l_1$. It follows the reference trajectory closely with minimal overshoot and fast convergence, particularly in the zoomed sections from 8 to 13 s and 23 to 28 s. In contrast, the PID controller has the largest tracking errors, especially during the transient phase. Considering the MPC-HG, this method exhibits a generally accurate tracking response compared to PID and SMC-HG, but still exhibits a small overshoot compared to LMPC-HG. Meanwhile, LMPC-HG continuously displays reduced and more stable errors throughout the simulation, which is further supported by the tracking error plots $e_x$ and $e_{l_1}$. PID exhibits the biggest and longest-lasting faults, while SMC-HG oscillates more. In Fig. 4, showing the swing angles $\phi_1$ (hook) and $\phi_2$ (load), all methods maintain small swing angles throughout the simulation. Especially, the LMPC-HG controller exhibits slightly better damping, particularly in $\phi_1$, where it has smaller oscillations overall.

Figure 5 illustrates the control input signals $F_x$ and $F_l$ for the PID, SMC-HG, MPC-HG, and proposed LMPC-HG controllers. It is evident that the SMC-HG and PID control signals rise sharply in the first stage, necessitating quick system modifications that provide difficulties for the actuators. While the SMC-HG reduces oscillations, it exhibits noticeable chattering, especially in $F_{l_1}$, which could negatively impact actuator longevity. Meanwhile, the MPC-HG controller demonstrates relatively smooth and bounded control signals, with moderate overshoot at the initial transient period. The LMPC-HG, however, provides smoother control inputs with minimal oscillations, reflecting its ability to generate optimal control actions while maintaining system stability and reducing the risk of actuator wear.

The lump disturbances $\mathcal{L}_j$, $(j = 1, 2, 3, 4)$ and their estimated values $\hat{\mathcal{L}}_j$ are depicted in Fig. 6, while the estimation errors of the lumped disturbances are shown in Fig. 7. Most errors converge close to zero, with some oscillations observed in $\tilde{\mathcal{L}}_2$ and $\tilde{\mathcal{L}}_4$, but these oscillations diminish over time. Overall, the high-gain observer offers precise disturbance estimation during the simulation.

To assess the effectiveness of the proposed approach, both qualitative and quantitative comparisons of tracking performance between the four approaches are carried out. In this study, we evaluate tracking error $\tilde{\boldsymbol{q}}_a = [e_1, e_2]^{\mathrm{T}}$ and the vibrant error $\tilde{\boldsymbol{q}}_u = [e_3, e_4]^{\mathrm{T}}$ according to four performance indexes as follows: integral squared error (ISE), integral time-multiplied squared error (ITSE), integral absolute error (IAE), and integral time-multiplied absolute error (ITAE). These evaluation results are shown in Table 2.

It is evident from the performance indices in Table 2 that the suggested control approach considerably enhances the tracking and vibration indices in comparison with the remaining three approaches.

Specifically, for the ISE index, which penalizes larger errors more than smaller ones, the LMPC-HG controller achieves the best performance with the smallest $\text{ISE}_{\boldsymbol{q}_a}$ value of 2.74e−04, slightly lower than 5.63e−04 of the MPC-HG method and outperforms SMC-HG (0.0028) and PID (0.0051). This indicates its strong capability in reducing tracking error during the transient phase. A similar trend is observed for $\text{ISE}\boldsymbol{q}_u$, which shows the ability to reduce vibration, LMPC-HG (0.0658) and MPC-HG (0.0660) both demonstrate better performance than SMC-HG (0.0694) and PID (0.0689).

Regarding the ITSE index, which places greater weight on errors occurring at later times, the proposed method continues to provide the lowest $\text{ITSE}_{\boldsymbol{q}_a}$ (0.0035), suggesting effective long-term tracking performance. Upon following this, MPC-HG ranks second with a value of 0.0042, again bet-

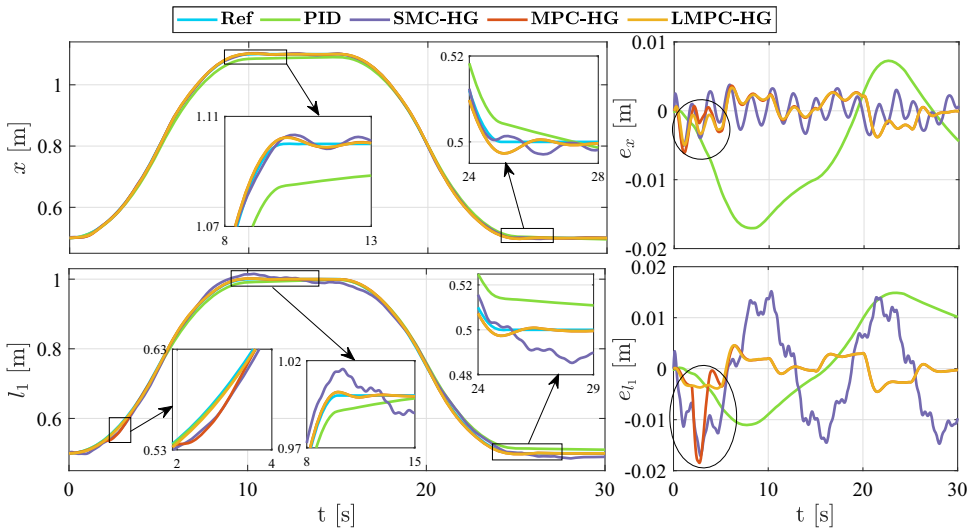



**Fig. 3** Output signals $x$ $[m]$, $l_1$ [m] and their tracking errors in scenario 1

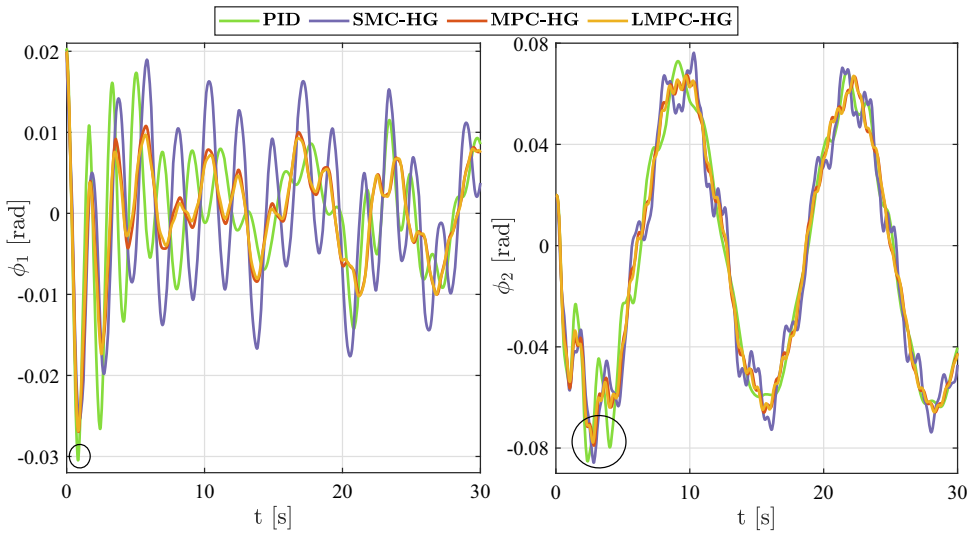



**Fig. 4** Swing angles of the hook $\phi_1$ and the load $\phi_2$ in scenario 1

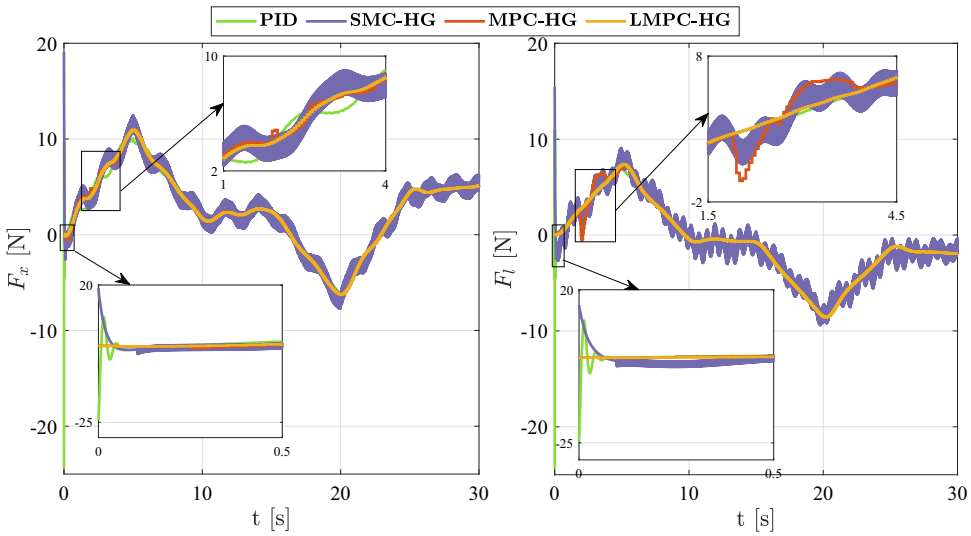



**Fig. 5** The driving forces acting on the system $F_x$ and $F_{l_1}$ in scenario 1

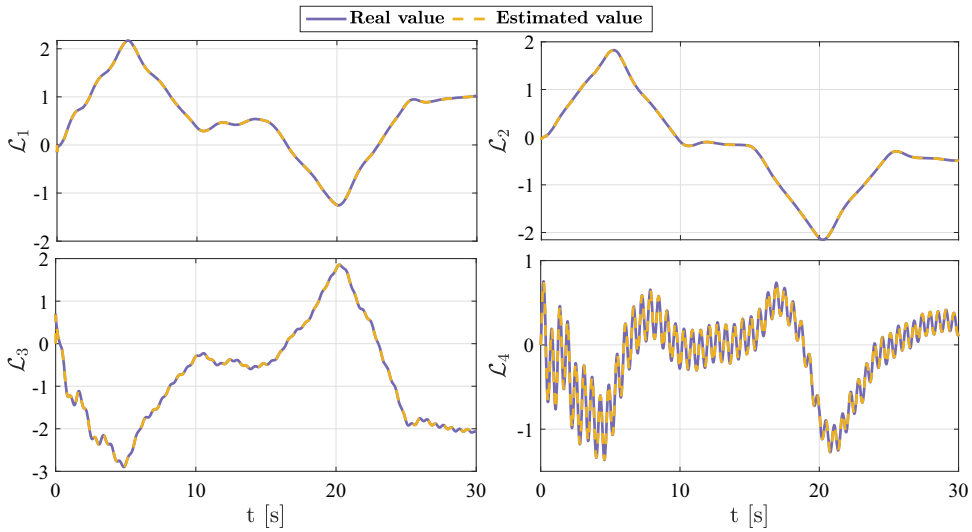



**Fig. 6** The lump disturbance $\mathcal{L}_j$ and their estimated values $\hat{\mathcal{L}}_j$ in scenario 1

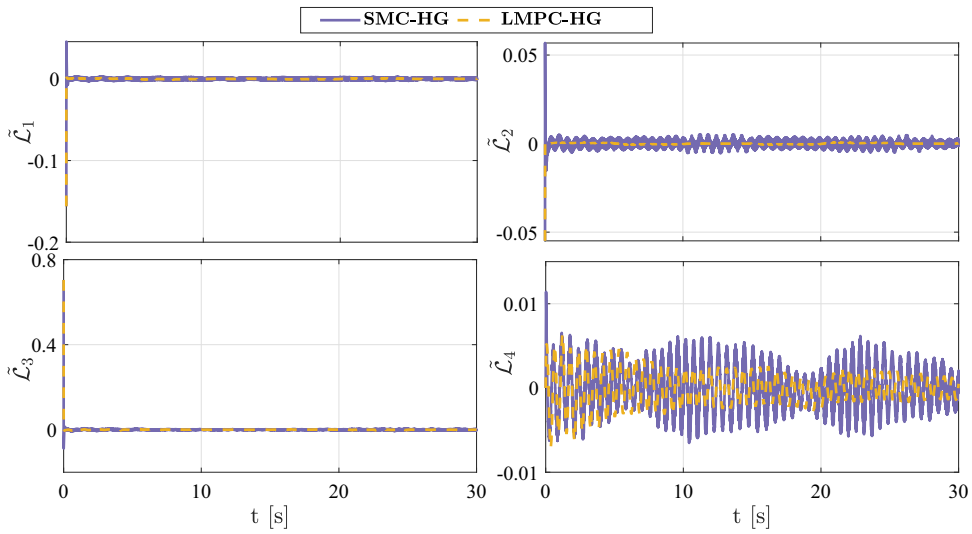



**Fig. 7** The estimation errors of the lump disturbance $\tilde{\mathcal{L}}_j$ in scenario 1

ter than SMC-HG (0.0410) and PID (0.0771). For ITSE$\boldsymbol{q}_u$, both LMPC-HG (0.9754) and MPC-HG (0.9763) outperform SMC-HG (1.0263) and PID (1.0095), with minimal difference between LMPC-HG and MPC-HG.

For the IAE index, which accumulates the total absolute error, LMPC-HG achieves the lowest values for both IAE$_{\boldsymbol{q}_a}$ (0.1058) and IAE$\boldsymbol{q}_u$ (1.4048), indicating excellent steady-state accuracy. Meanwhile, MPC-HG shows slightly worse but still competitive results (0.1177 for $\boldsymbol{q}_a$ and 1.4116 for $\boldsymbol{q}_u$), better than those of SMC-HG (0.3010 and 1.5096) and PID (0.4625 and 1.4859).

Finally, in terms of the ITAE index, which heavily penalizes persistent errors, LMPC-HG again leads with the lowest ITAE$_{\boldsymbol{q}_a}$ value of 1.4168. MPC-HG ranks second (1.4476), followed by SMC-HG (4.4763) and PID (7.1163). Regarding ITAE$_{\boldsymbol{q}_u}$, LMPC-HG achieves the smallest value (20.9165), slightly ahead of MPC-HG (20.9806), both of which outperform SMC-HG (22.4003) and PID (21.9035).

In summary, for the double-pendulum crane system under disturbances, the LMPC-HG approach offers the highest performance across all indices, including more accuracy and better vibration suppression.

## 4.2 Scenario 2

In this scenario, to further evaluate the novelty and effectiveness of the proposed method under different conditions, a different type of external disturbance is employed. The corresponding disturbance signal is illustrated in Fig. 8.

Figure 9 presents the tracking responses of the four controllers under white noise disturbance. Both MPC-HG and LMPC-HG achieve nearly identical tracking accuracy, following the reference closely with minimal overshoot and rapid convergence. These results confirm the robustness of predictive controllers based on high-gain disturbance observers in handling noisy environments. In contrast, PID

**Table 2** Comparison of control strategies

| States | Control strategy | Performance indices ISE | ITSE | IAE | ITAE |
|---|---|---|---|---|---|
| $q_a$ | LMPC-HG | **2.74e−04** | **0.0035** | **0.1058** | **1.4168** |
| | MPC-HG | 5.63e−04 | 0.0042 | 0.1177 | 1.4476 |
| | SMC-HG | 0.0028 | 0.0410 | 0.3010 | 4.4763 |
| | PID | 0.0051 | 0.0771 | 0.4625 | 7.1163 |
| $q_u$ | LMPC-HG | **0.0658** | **0.9754** | **1.4048** | **20.9165** |
| | MPC-HG | 0.0660 | 0.9763 | 1.4116 | 20.9806 |
| | SMC-HG | 0.0694 | 1.0263 | 1.5096 | 22.4003 |
| | PID | 0.0689 | 1.0095 | 1.4859 | 21.9035 |

and SMC-HG show significant performance degradation. Notably, the PID controller struggles to track the hook length $l_1$, with large steady-state error and sluggish convergence, especially during periods of disturbance. The SMC-HG controller, although better than PID, exhibits high-frequency oscillations due to its limited noise rejection capability. The superiority of LMPC-HG and MPC-HG in noise rejection is further confirmed by the error plots $e_x$ and $e_{l_1}$, where both controllers maintain low and stable errors, whereas PID and SMC-HG show large fluctuations and delayed settling. Particularly, SMC-HG produces high-frequency oscillations due to its limited robustness against noisy signals, which reduces its practical applicability.

In Fig. 10, the swing angles $\phi_1$ and $\phi_2$ are plotted to evaluate the anti-sway capability. Although all controllers manage to keep the swing within acceptable ranges, LMPC-HG demonstrates slightly better damping characteristics, especially for $\phi_2$ at later time intervals. Notably, both PID and SMC-HG suffer from persistent oscillations with higher amplitudes, while LMPC-HG and MPC-HG achieve better suppression. However, LMPC-HG still outperforms MPC-HG in maintaining lower and more stable swing magnitudes over time.

Figure 11 illustrates the control input signals $F_x$ and $F_l$ for four controllers under white noise disturbance. As seen in the figure, both PID and SMC-HG produce control inputs with sharp rises in the initial stage, particularly in $F_x$, which can lead to actuator stress and reduced hardware lifespan. Furthermore, SMC-HG also suffers from high-frequency chattering, most notably in $F_{l_1}$, which could degrade actuator performance over time. Although the control signals of MPC-HG and LMPC-HG exhibit a step-wise behavior due to the predictive nature of the control update, they still maintain bounded and relatively less oscillatory responses compared to SMC-HG. Both controllers provide more consistent control magnitudes with fewer high-frequency fluctuations, especially after the transient period. This suggests that while MPC-based methods do not produce smooth signals in the strict sense, they offer better robustness and actuator-friendliness under disturbance conditions than conventional PID or sliding mode approaches.

Figure 12 presents the estimation performance of the HGDO for the total lumped disturbances $\mathcal{L}_1$ to $\mathcal{L}_4$ under external white noise excitation. As shown, the estimated signals (dashed yellow) closely follow the actual disturbance profiles (solid blue) in all channels, demonstrating accurate and real-time tracking. The observer achieves a high degree of matching not only in low-frequency components but also in fast-varying segments of the signals. Even in $\mathcal{L}_4$, where fluctuations are more pronounced, the HGDO successfully captures the overall trend and amplitude. These results validate the robustness of the HGDO in handling high-frequency, stochastic disturbances and confirm its suitability for use in noisy environments.

## 5 Conclusion

This study introduced a Lyapunov-based model predictive control (LMPC) approach, integrated with a high-gain disturbance observer, for controlling a 2D double-pendulum overhead crane system. By employing the high-gain observer, disturbances and system uncertainties are effectively estimated, giving precise information to the controller. The LMPC provides precise reference trajectory monitoring while avoiding oscillations during crane action. Additionally, the control inputs remain within the drivers capacity, and the output signals, including two angles, stay within acceptable limits. The closed-loop system's global stability is obtained by using stability requirements derived from the Lyapunov function and second-order sliding mode control. Simulations and comparisons to conventional approaches indicate the effectiveness of the proposed control strategy. However, this control scheme needs the exact information about the inertia matrix, which may vary in practice due to changes in payload. In the future, the proposed control scheme will be tested in an experimental environment to validate its practical feasibility. Moreover, further research will focus on extending

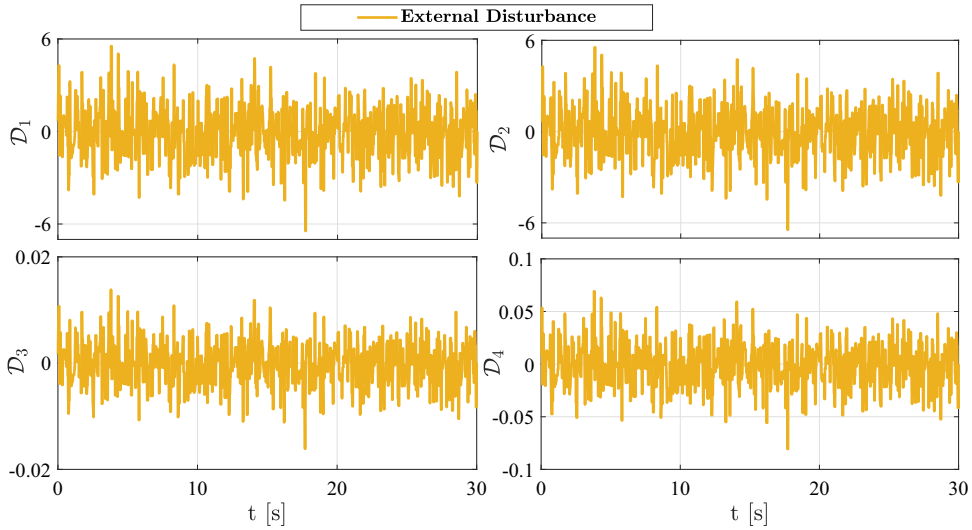



**Fig. 8** The external disturbance $\mathcal{D}_j$ in scenario 2

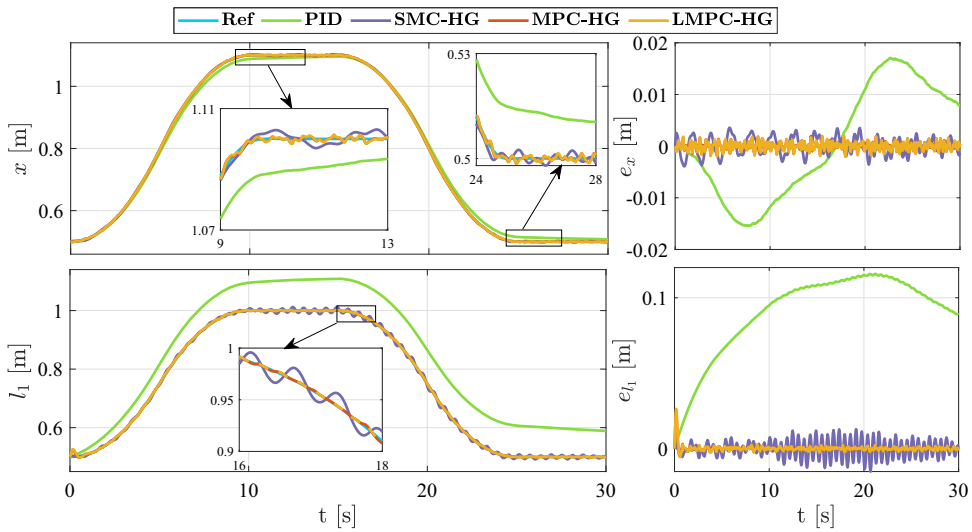



**Fig. 9** Output signals $x$ [$m$], $l_1$ [m] and their tracking errors in scenario 2

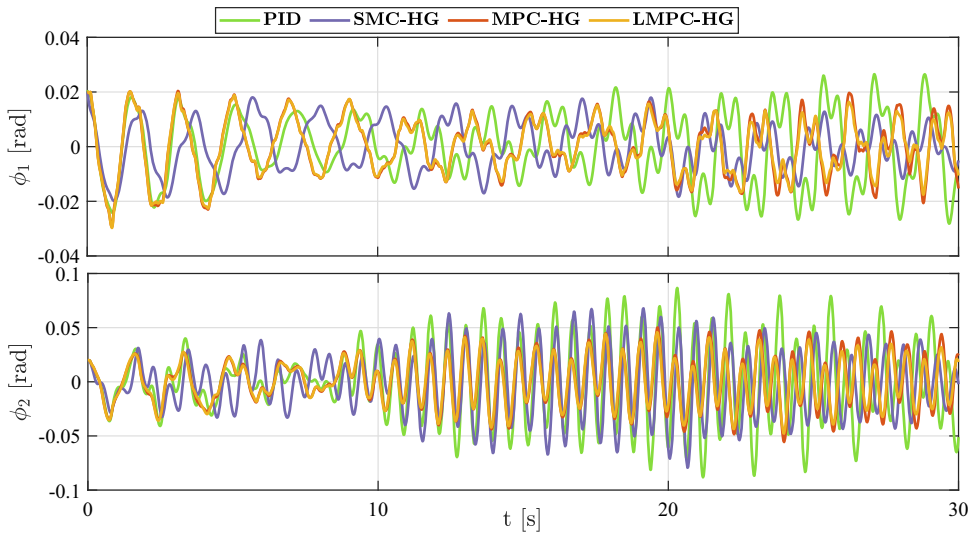



**Fig. 10** Swing angles of the hook $\phi_1$ and the load $\phi_2$ in scenario 2

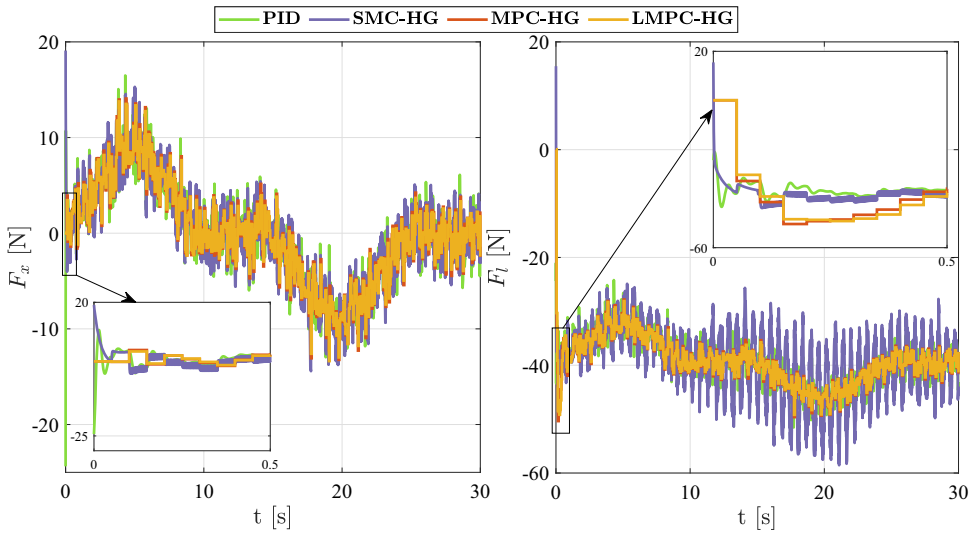



**Fig. 11** The driving forces acting on the system $F_x$ and $F_{l_1}$ in scenario 2

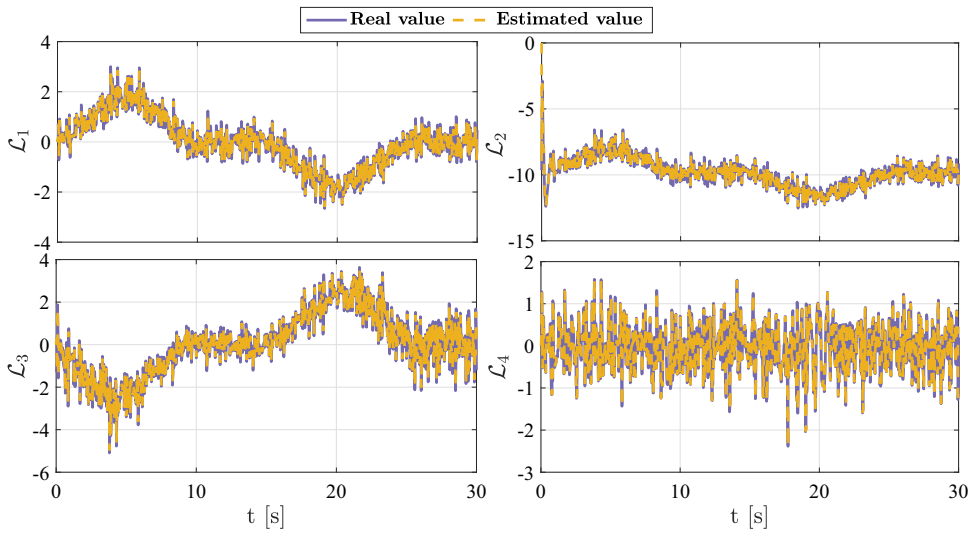



**Fig. 12** The lump disturbance $\mathcal{L}_j$ and their estimated values $\hat{\mathcal{L}}_j$ in scenario 2

the framework to account for time-delay effects and reducing the dependence on the exact knowledge of the system inertia matrix, thereby enhancing its robustness and applicability in real-world scenarios.

## Appendix

Matrices $\mathcal{M}(\boldsymbol{q}), \mathcal{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}), \mathcal{G}(\boldsymbol{q})$ are represented as the following formulas:

$$\mathcal{M}(\boldsymbol{q}) = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \\ m_{41} & m_{42} & m_{43} & m_{44} \end{bmatrix},$$

$$\mathcal{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \\ c_{41} & c_{42} & c_{43} & c_{44} v \end{bmatrix} \mathcal{G}(\boldsymbol{q}) = \begin{bmatrix} g_{11} & g_{21} & g_{31} & g_{41} \end{bmatrix}^{\mathrm{T}}.$$

The components of the mass matrix $\mathcal{M}(\boldsymbol{q})$ are expressed as

$m_{11} = M + m_1 + m_2, m_{12} = m_{21} = (m_1 + m_2)\sin\phi_1, m_{13} = m_{31} = (m_1 + m_2)l_1\cos\phi_1, m_{14} = m_{41} = m_2 l_2 \cos\phi_2, m_{22} = m_1 + m_2, m_{23} = m_{32} = 0, m_{24} = m_{42} = m_2 l_2 \sin(\phi_1 - \phi_2), m_{33} = (m_1 + m_2)l_1^2, m_{34} = m_{43} = m_2 l_1 l_2 \cos(\phi_1 - \phi_2), m_{44} = m_2 l_2^2.$

The elements of the $\mathcal{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ matrix are defined as:

$c_{11} = f_x, c_{12} = 2(m_1 + m_2)\cos\phi_1 \dot{\phi}_1, c_{13} = -(m_1 + m_2)l_1 \sin\phi_1 \dot{\phi}_1, c_{14} = -m_2 l_2 \sin\phi_2 \dot{\phi}_2, c_{21} = c_{31} = c_{33} = c_{41} = c_{44} = 0, c_{22} = f_l, c_{23} = -(m_1 + m_2)l_1\dot{\phi}_1, c_{24} = -m_2 l_2 \cos(\phi_1 - \phi_2)\dot{\phi}_2, c_{32} = 2(m_1 + m_2)l_1\dot{\phi}_1, c_{34} = m_2 l_1 l_2 \sin(\phi_1 - \phi_2)\dot{\phi}_2, c_{42} = 2m_2 l_2 \cos(\phi_1 - \phi_2)\dot{\phi}_1, c_{43} = -m_2 l_1 l_2 \sin(\phi_1 - \phi_2)\dot{\phi}_1.$

The coefficients of friction are denoted as $f_x$ and $f_l$, while the nonzero components of the gravity vector $\mathcal{G}(\boldsymbol{q})$ are expressed as $g_{11} = 0, g_{21} = (m_1 + m_2)g(-\cos\phi_1), g_{31} = (m_1 + m_2)g l_1 \sin\phi_1, g_{41} = m_2 l_2 g \sin\phi_2.$

Details of matrices $\mathcal{M}_{aa}(\boldsymbol{q})$, $\mathcal{M}_{au}(\boldsymbol{q})$, $\mathcal{M}_{ua}(\boldsymbol{q})$, and $\mathcal{M}_{uu}(\boldsymbol{q})$ are as follows: $\mathcal{M}_{aa}(\boldsymbol{q}) = \begin{bmatrix} m_{11}\ m_{12} \\ m_{21}\ m_{22} \end{bmatrix}$, $\mathcal{M}_{au}(\boldsymbol{q}) = \mathcal{M}_{ua}^{\mathrm{T}}(\boldsymbol{q}) = \begin{bmatrix} m_{13}\ m_{14} \\ m_{23}\ m_{24} \end{bmatrix}$ $\mathcal{M}_{uu}(\boldsymbol{q}) = \begin{bmatrix} m_{33}\ m_{34} \\ m_{43}\ m_{44} \end{bmatrix}$.

**Author Contributions** All the authors performed the calculations and simulations, analyzed the data, discussed the results and contributed to the final manuscript.

## Declarations

**Conflict of interest** The authors declare that they have no conflict of interest.

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